

6.3 Classwork: Linear and Quadratic Functions — Markscheme

1. A straight line L passes through the points $A(-2, 5)$ and $B(4, -1)$.

- (a) Find the gradient of L . [2]

$$m = \frac{-1 - 5}{4 - (-2)} = \frac{-6}{6} \quad (\text{M1})$$

$$= -1 \quad (\text{A1})$$

- (b) Find the equation of L in the form $y = mx + c$. [2]

$$y - 5 = -1(x - (-2)) \quad (\text{M1})$$

$$y = -x + 3 \quad (\text{A1})$$

- (c) Find the x -intercept of L . [2]

$$0 = -x + 3 \quad (\text{M1})$$

$$x = 3 \Rightarrow (3, 0) \quad (\text{A1})$$

- (d) A second line L_2 is parallel to L and passes through the point $(0, 7)$. Find the equation of L_2 . [2]

$$\text{Parallel} \Rightarrow m = -1 \quad (\text{M1})$$

$$y = -x + 7 \quad (\text{A1})$$

2. Let $f(x) = x^2 - 4x - 12$.

(a) Factorise $f(x)$. [2]

$$f(x) = (x - 6)(x + 2) \quad (\mathbf{A1})(\mathbf{A1})$$

(b) Hence find the x -intercepts of the graph of f . [1]

$$x = 6 \text{ and } x = -2 \quad (\mathbf{A1})$$

(c) Find the coordinates of the vertex of the parabola. [3]

$$x = \frac{-(-4)}{2(1)} = 2 \quad (\mathbf{M1})$$

$$f(2) = 4 - 8 - 12 = -16 \quad (\mathbf{A1})$$

$$\text{Vertex} = (2, -16) \quad (\mathbf{A1})$$

(d) Write down the y -intercept. [1]

$$f(0) = -12 \Rightarrow (0, -12) \quad (\mathbf{A1})$$

(e) Write down the range of f . [1]

$$f(x) \geq -16 \quad (\mathbf{A1})$$

3. The line L_1 has equation $2x + y - 8 = 0$.

(a) Write the equation of L_1 in the form $y = mx + c$. [2]

$$2x + y - 8 = 0 \quad (\text{M1})$$

$$y = -2x + 8 \quad (\text{A1})$$

(b) Write down the gradient and the y -intercept of L_1 . [1]

$$\text{Gradient} = -2, \quad y\text{-intercept} = 8 \quad (\text{A1})$$

(c) The line L_2 is perpendicular to L_1 and passes through the point $(2, -1)$. Find the equation of L_2 in the form $ax + by + d = 0$, where $a, b, d \in \mathbb{Z}$. [3]

$$\text{Perpendicular gradient: } m_2 = \frac{1}{2} \quad (\text{A1})$$

$$y - (-1) = \frac{1}{2}(x - 2) \quad (\text{M1})$$

$$2y + 2 = x - 2 \quad (\text{A1})$$

$$x - 2y - 4 = 0$$

(d) Find the coordinates of the point of intersection of L_1 and L_2 . [3]

$$-2x + 8 = \frac{1}{2}x - 2 \quad (\text{M1})$$

$$-4x + 16 = x - 4 \quad (\text{A1})$$

$$5x = 20 \Rightarrow x = 4, \quad y = -2(4) + 8 = 0$$

$$\text{Point of intersection: } (4, 0) \quad (\text{A1})$$

4. Let $g(x) = 2x^2 - 8x + 6$.

(a) Write $g(x)$ in the form $a(x - h)^2 + k$. [3]

$$g(x) = 2(x^2 - 4x) + 6 \quad (\text{M1})$$

$$= 2(x^2 - 4x + 4) - 8 + 6 \quad (\text{A1})$$

$$= 2(x - 2)^2 - 2 \quad (\text{A1})$$

(b) Write down the coordinates of the vertex of the parabola. [1]

$$\text{Vertex} = (2, -2) \quad (\text{A1})$$

(c) Find the x -intercepts of the graph of g . [2]

$$2(x - 2)^2 - 2 = 0 \quad (\text{M1})$$

$$(x - 2)^2 = 1 \Rightarrow x - 2 = \pm 1$$

$$x = 1 \text{ and } x = 3 \quad (\text{A1})$$

(d) The graph of g is reflected in the x -axis. Write the equation of the new graph in the form $y = ax^2 + bx + c$. [2]

$$y = -g(x) = -(2x^2 - 8x + 6) \quad (\text{M1})$$

$$y = -2x^2 + 8x - 6 \quad (\text{A1})$$